(Communication Systems) ECC - 209

(Analog Communication) [UPMA SINGH]

Communication :- Communication is the process of establishing connection or link b/w two points for information exchange.

or

Communication is simplay the basic process of exchanging information.

Electronic equipments;-

- Electronic equipments which are used for communication purpose, are called Communication equipments.
- Different communication equipments when assembled together form a communication system.
  e.g. radia bradcasting, point-to-point communication, mobile communication, computer communication, radar communication, television broadcasting, radio telemetry etc.

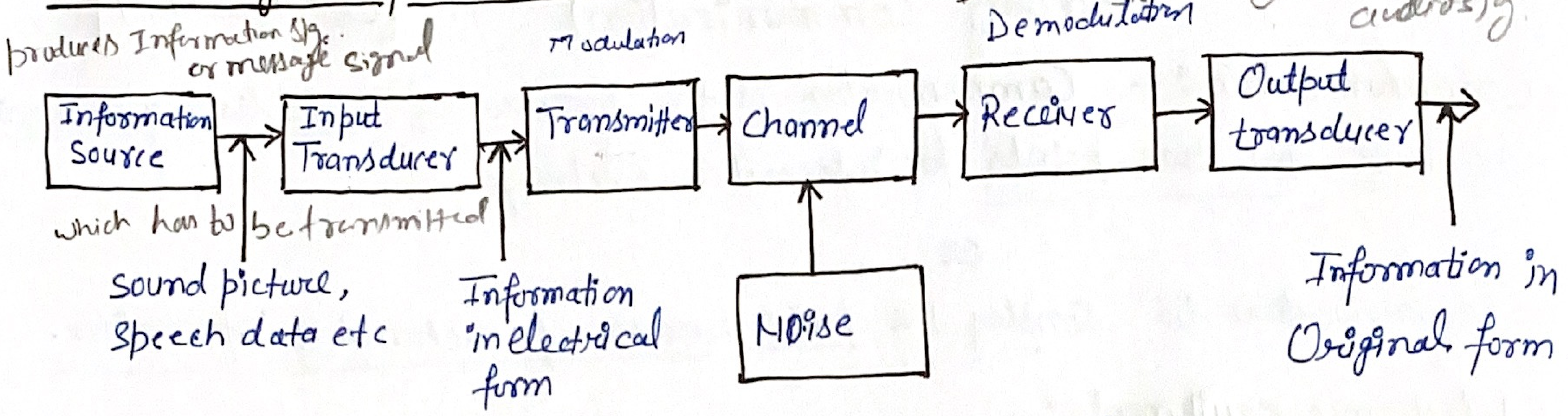
The communication Process : Elements of a Communication System.

- The fundamental purpose of a communication system is to transfer information from one place to another.
- So electronic communication involves three steps
  1. Transmission
  2. Reception
  3. processing of information b/w two or more locations using electronic circuits.

- The original source of information can be in analog form, such as human voice or music, or in digital form, such as binary-coded numbers or alphanumeric codes.

" So, the purpose of a communication system is to transmit an information bearing s/g, from a source, located at one point, to a destination, located at another point some distance away."

Block diagram of a communication System :-



→ The main function of comm. system is to communicate a message.

1- Information Source

The function of information source is to produce required message which has to be transmitted.

In general, there can be various messages in the form of words, group of words, code, symbols, sound signal etc. However, out of these messages, only the desired message is selected and conveyed or communicated.

2- Input Transducer

A transducer is a device which converts one form of energy into another form. The message from the information source may or may not be electrical in nature, an input transducer is used to convert it into a time-varying electrical signal.

e.g. in case of radio-broadcasting, a microphone converts the information or message which is in the form of sound waves into corresponding electrical signal.

3- Transmitter

The function of the transmitter is to process the electrical s/g from different aspects.

e.g. in radio broadcasting the electrical s/g obtained from sound signal, is processed to restrict its range of audio frequencies (upto 5kHz in amplitude modulation radio broadcast). and is often amplified. In wire telephony, no real processing is needed. However, in long-distance radio comm. or broadcast, signal ampli-fication is necessary before modulation.

→ Modulation is the main function of the transmitter

In modulation, the message signal is superimposed upon the high-frequency carrier signal.

→ In short, we can say that inside the transmitter, signal processing such as restriction of range of audio frequencies, amplification and modulation are achieved.

4. The channel and the Noise

→ Channel means the medium through which the message travels from the transmitter to the receiver.

or

→ channel provides the physical connection b/w the transmitter and the receiver.

→ There are two tyes of channels:-

1) point-to-point channels → Wire lines, microwave links, optical fibres.
2) broadcast channels.

Wire lines, operate by guided electromagnetic waves and they are used for local telephone transmission.

Microwave links are used in long-distance telephone transmission

Optical fibres are used in optical communications

↳ optical fiber is a low-loss, well-controlled, guided optical medium.

→ They all provide a physical medium for the transmission of signals from one point to another point. Therefore, for these channels, the term point-to-point is used.

One the other hand, the broadcast channels provide a capability where several receiving stations can be reached simultaneously from a single transmitter.

e.g satellite

→ During the process of transmission and reception the signal gets distorted due to noise introduced in the system. Noise may interfere with signal at any point in a communication system.

5- Receiver

The main funⁿ of the receiver is to reproduce the message signal in electrical form from the distorted received signal. This reproduction of the original signal is accomplished by a process known as the demodulation or detection.

→ Demodulation is the reverse process of modulation carried out in transmitter.

6- Destination

It is the final stage which is used to convert an electrical message s/g into its original form.

e.g. in radio broadcasting, the destination is a loudspeaker which works as a transducer ie.

⇒ it converts the electrical s/g in the form of original sound signal.

★ Bandwidth

· Difference b/w the upper and lower frequency limits of the signal is called Bandwidth.

or

freq range of a signal is known as it B.W

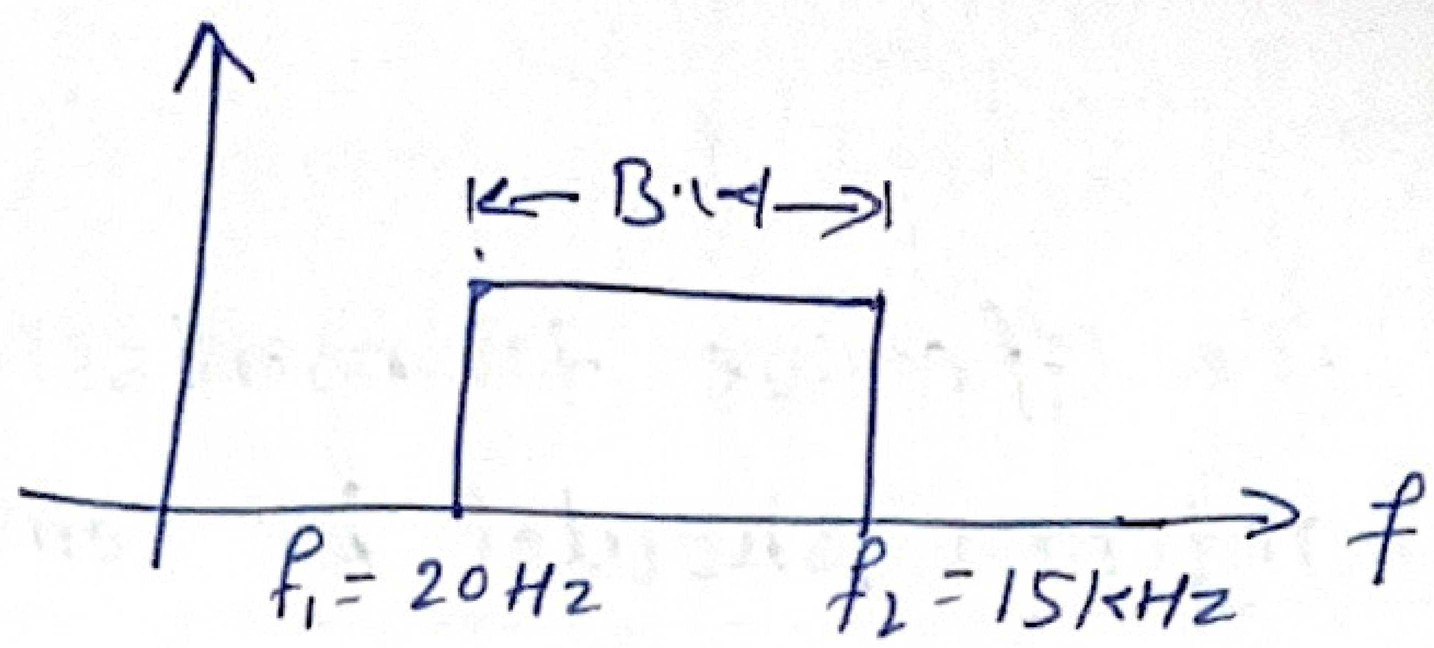
There are different types of passband signals.

1) voice signal
2) music signal
3) TV signal

Each of these signals will have its own frequency range.

e.g. music signal. range is 20 Hz to 15 KHz

$$BW = f_2 - f_2$$
$$= 15000 - 20$$
$$= 14980\ Hz$$



S.No	Type of signal	Range of frequency in Hz	Bandwidth in Hz.
1.	Voice signal (speech) for telephony	300 – 3400	3,100
2.	Music signal	20 – 15000	14,980
3	TV signals	0 – 5MHz 30MHz - 300MHz	5MHz
4	Digital data	300 – 3400 If it is using the telephone line for its transmission.	3,100

Audio freq range = 20 Hz – 20KHz

music signal = 20Hz – 15KHz

Modulation:– Modulation is defined as the process by which some characteristics (i.e amplitude, frequency, and phase) of a carrier wave are varied in accordance with a modulating wave.

Demodulation:– Demodulation is the reverse process of modulation which is used to get back the original message signal.

→ Modulation is performed at the transmitting end whereas demodulation is performed at the receiving end.

→ In analog modulation sinusoidal signal is used as carrier where as in digital modulation pulse train is used as carrier.

1- Reduction of Noise.

Need for Modulation

1- Practicability of antenna (Practical length of Antenna)

For the effective transmission of a signal, the height h of the antenna should be comparable to the wavelength λ of the signal, at least the height of the antenna h should be $\lambda/4$ in length so that the antenna can sense the variations of the signal properly.

→ The low-frequency message signal has a very high value of λ which will require a very high antenna (practically not possible)

Ex If we have to transmit a signal of 20 kHz, then $\lambda = c/f$ and height of the antenna $h \approx \lambda$ where c is the wave velocity, here $c = 3\times10^8$ m/s.

Ans $h \approx \lambda = (3\times10^8)/(20\times10^3)$ $\qquad \lambda = \frac{c}{f} = \frac{3\times10^8}{20\times10^3}$

$h = 15$ km.

Hence, we need to modulate the message signal over the high-frequency carrier signal so that we can have a practical value for the height h of the antenna.

2- Narrow Banding of Signal

An audio signal usually has a frequency range (20 Hz to 20 KHz), if it is directly transmitted then the ratio of highest to the lowest frequency becomes (20 KHz / 20 Hz) = 1000.

But if this audio signal is modulated over a carrier signal of frequency 1000 kHz then the ratio of highest to the lowest frequency becomes.

$$\frac{(1000\,\text{kHz} + 20\,\text{kHz})}{(1000\,\text{kHz} + 20\,\text{Hz})} \cong 1.2$$

Hence, we need modulation to convert a wideband signal into a narrow band signal.

3- Effective Power Radiated by Antenna.

Power radiated by an antenna $\propto (l/\lambda)^2$ where l is the length of the antenna and λ is the wavelength of the signal which is to be transmitted through the antenna. This relation clearly shows that when signals having a low frequency and high wavelength is transmitted directly the power radiated by the antenna is very low and the signal will vanish after travelling some distance.

low freq
Hence, to transmit such signals over long distances, we superimpose these low-frequency signals over the carrier signal having a high frequency and short wavelength so that the power radiated by the antenna of the same length will be very large.

4- Noise Reduction:-

Noise is the major limitation of any communication. Although noise can not be eliminated completely, with the help of several modulation schemes, the effect of noise can be minimized.

5- Multiplexing is possible:-

Multiplexing is a process in which two or more signals can be transmitted over the same communication channel simultaneously. This is possible only with modulation. The multiplexing allows the same channel to be used by many signals. Hence, many TV channels can use the same frequency range, without getting mixed with each other or different frequency signals can be transmitted at the same time.

Types of Modulation

→ Modulation is basically of two types:

1- Continuous wave modulation

2- Pulse modulation

1- Continuous wave modulation:-

- When the carrier wave is continuous in ~~ed~~ nature, the modulation process is known as continuous wave (CW) modulation or analog modulation.

- e.g. 1) Amplitude modulation
  2) Angle modulation

Amplitude modulation :-

When the amplitude of the carrier sig is varied, in accordance with the message signal, it is known as amplitude modulation (AM).

Angle modulation :-

When the angle of the carrier signal is varied, in accordance with the message signal, it is known as angle modulation.

→ Angle modulation may be further subdivided into;

1) Frequency modulation (FM) 2) Phase modulation (PM)

Frequency modulation :-

When the instantaneous frequency of the carrier signal is varied, in accordance with the message signal, it is known as frequency modulation.

Phase modulation :-

When the instantaneous phase of the carrier signal is varied, in accordance with the message signal, it is known as phase modulation.

2- Pulse Modulation :-

- When the carrier wave is a pulse-type waveform, the modulation process is known as pulse modulation.
- In pulse modulation, the carrier consists of a periodic sequence of rectangular pulses.
- Pulse modulation can be of an analog or digital type.
- In Analog pulse modulation, the amplitude, width and position of a pulse is varied in accordance with sample value of the message signal.
- The analog pulse modulation may be of the following three types:
  (i) Pulse amplitude modulation (PAM)
  (ii) Pulse width modulation (PWM)
  (iii) Pulse position modulation (PPM)

⇒ Whereas, the digital form of pulse modulation is known as pulse-code modulation (PCM).

- Modulation
  - Continuous wave modulation
    - Amplitude modulation
      - DSB-SC
      - DM SSB
      - VSB
    - Angle modulation
      - Frequency modulation
        - Wideband WBFM
        - Narrowband NBFM
      - Phase modulation
  - Pulse modulation
    - Pulse analog modulation
      - PAM (Pulse amplitude Modulation)
      - PTM
        - PWM/PDM/PLM
        - PPM
    - Pulse digital Modulation

PTM Pulse time Modulation
PLM = Pulsed load Modulation
PDM = Pulse density Modulation
PWM = Pulse width Modulation
PPM = Pulse position Modulation